Assignment : 16

Problem Statement: To accept a list of N integers & partition the list into two sublists containing even and odd numbers.

Objectives:

- To understand how to work with arrays in C.
- To learn how to separate numbers based on conditions.
- To practice the use of loops and conditional statements.
- To display two different lists for even & odd numbers.

Theory: In C programming, arrays are used to store multiple values of the same datatype. By using loops & and conditional statements, It is possible to examine each element in the array and classify it based on certain conditions. In this program, the user enters N integers which are stored in an array. Each number is then ~~co~~ checked using the modulus operator (%). If a number is divisible by 2, it is categorized as an even number, otherwise, it is an odd number.

Platform: Linux

Input: Number of integers (N) and the list of integers.

Output: Displays two lists : Even numbers & Odd numbers.

Sample Input: Enter number of elements : 5
Enter integers : 16 7 4 9 12

Sample Output: Even numbers are: 10 4 12
Odd numbers are: 7 9

Conclusion: Thus, the C program successfully accepts a list of integers and partitions them into two separate lists of even & odd numbers.

FAQs:

1) Which operator is used to check even and odd numbers in C?

A. a) In C, the modulus operator % is used to check whether a number is even or odd.

b) Ex:

```
if (num % 2 == 0)
    printf("Even");
else
    printf("odd");
```

2) Why are arrays used in this program?

A. a) Efficient Storage: Instead of writing num1, num2, num3, ..., you can use one array like num[].

b) Easy processing: You can use loops to access & process all values.

c) Less code, more clarity: Makes the program shorter & easier to understand.

3) How does the modulus (%) operator help in separating even and odd numbers?

A. a) The modulus operator % gives the remainder when one number is divided by another. This is what helps us identify even & odd numbers.